BUS-F420: Equity and Fixed Income Investments

Lecture 13: Measuring Bond Yields

Prof. Derek Wenning

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Recap from last time

- Last time, talked about how to value a bond

- **Key mechanisms:**
  1. Lower maturities $\rightarrow$ higher prices *if selling at a discount*
  2. Embedded (call) options $\rightarrow$ lower prices to compensate
  3. Higher market yields $\rightarrow$ lower prices to compensate

- **Today:** Given the **price** of a bond, what is its implied **yield**?
  $\rightarrow$ Closer to an idea of its *return* given its *cost*

Why/when do we care about prices vs. yields?

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- Although the two are intimately connected, they serve different purposes

- **Prices** reflect value, which matters for *asset holdings*
  - Ex 1: Banks must report *risk-based capital*, which is derived from asset valuations
  - Ex 2: Retirement accounts

- **Yields** reflect income relative to costs, which matters for *returns*
  - Allows us to make comparisons across relatively *dissimilar* bonds
  - **Today:** what information is contained in a yield?

What do prices tell us about yields?

- **Question:**
  If we hold a bond to maturity, what is the *yield* we should receive that is consistent with a given price?

- Call this the **yield to maturity**, which solves

$$\text{(Given!)} \rightarrow \boldsymbol{P} = \sum_{t=1}^{T} \frac{CF_t}{(1+\boldsymbol{y})^t} \leftarrow \text{(Needs to be determined!)}$$

Calculating Yield to Maturity and Bond Prices in Excel

- **Challenge:** We can't *directly* solve for **yield to maturity** $\rightarrow$ use Excel!

- **Key Function:** `RATE(periods, coupon, −price, par value)`
  - `periods`: # of payment periods remaining in the bond
  - `coupon`: size of the coupon payment ($r \times M$)
  - `price`: current price of the bond — **must be entered negatively**
  - `par value`: par value of the bond

Calculating Yield to Maturity and Bond Prices in Excel

- Let's work through an example using a 5-year corporate bond where:
  - (Annual!) Coupon rate (r): 8%
  - Price: \$900
  - Par value: \$1,000

- How should we fill out our Excel function?

Formula: `RATE(periods, coupon, -price, par value)`

Our bond: `RATE(`

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- How should we fill out our Excel function?

Formula: $\texttt{RATE}(\texttt{periods}, \texttt{coupon}, \texttt{-price}, \texttt{par value})$

Our bond: $\texttt{RATE}(10, 40, -900, 1000) = \mathbf{5.315\%}$

Let's practice with the following bonds:

`RATE(periods, coupon, -price, par value)`

Years Remaining	Annual Coupon Rate	Market Price	Par Value	**YTM**
8	10%	\$850	\$1,000	
6	10%	\$850	\$1,000	
8	**8%**	\$850	\$1,000	
8	10%	**\$800**	\$1,000	

(Remember to add decimal places to your computed rates!)

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8	**8%**	\$850	\$1,000	**5.43%**
8	10%	**\$800**	\$1,000	**7.14%**

(Remember to add decimal places to your computed rates!)

What information is imbedded in yield to maturity?

- **So far:** Two key characteristics determine yields — coupon rates + capital gains
- But let's rearrange our equation for yield to maturity:

$$\underbrace{(1+y)^T}_{\substack{\textbf{required return} \\ \textbf{(total yield to maturity)}}} =$$

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- **Imbeds a key assumption: reinvestment of coupon payments!**

How important is each channel?

$$(1+y)^T = \frac{TC}{P} + \frac{M}{P} + \frac{C\sum_{t=1}^{T} R_{T-t}(y)}{P}$$

Years Left	Annualized Coupon Rate	Market Price	Par Value	*Total* YTM	Current Yield	Capital Gains	Reinv. Yield
8	10%	\$850	\$1,000	175.5%			
6	10%	\$850	\$1,000	122%			
8	8%	\$850	\$1,000	133%			
8	10%	\$800	\$1,000	201%			

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8	10%	\$850	\$1,000	175.5%	94.1%		
6	10%	\$850	\$1,000	122%			
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6	10%	\$850	\$1,000	122%	70.6%	17.7%	33.9%
8	8%	\$850	\$1,000	133%	75.3%	17.7%	40.0%
8	10%	\$800	\$1,000	201%	100%	25%	76.3%

What are the limits to yield to maturity?

- Clearly, reinvestment is an important part of yields!
- **Q: What are some issues with how we've treated reinvestment income?**

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- Clearly, reinvestment is an important part of yields!
- **Q: What are some issues with how we've treated reinvestment income?**
  - Assumed a *consistent* reinvestment rate!
- **True yields will likely reflect reinvestment risk**
  - Depend on *how* and *when* you reinvest!

Portfolio Yields

- So far, we've talked about a **single** bond. But what about a **portfolio?**

  Suppose you hold the following bonds (market value: $2,617):

Bond	Years	Annual Coupon Rate	Market Price	Par Value	YTM
A	1	8%	$945.75	$1,000	7%
B	2	7%	$850.75	$1,000	8%
C	3	10%	$820.50	$1,000	9%

- Q: What do you think the yield to maturity of your portfolio should be?

Portfolio Yields

	Cash Flow				(Discount rate: 8.000%)
Period	Bond A	Bond B	Bond C	Total Cash Flow	Discounted Cash Flow
1					
2					
3					
4					
5					
6					

True portfolio value: 2617

Portfolio Yields

(Discount rate: 8.000%)

	Cash Flow				
Period	Bond A	Bond B	Bond C	Total Cash Flow	Discounted Cash Flow
1	40				
2	1040				
3	0				
4	0				
5	0				
6	0				

True portfolio value: 2617

Portfolio Yields

(Discount rate: 8.000%)

	Cash Flow				
Period	Bond A	Bond B	Bond C	Total Cash Flow	Discounted Cash Flow
1	40	35			
2	1040	35			
3	0	35			
4	0	1035			
5	0	0			
6	0	0			

True portfolio value: 2617

Portfolio Yields

(Discount rate: 8.000%)

	Cash Flow				
Period	Bond A	Bond B	Bond C	Total Cash Flow	Discounted Cash Flow
1	40	35	50		
2	1040	35	50		
3	0	35	50		
4	0	1035	50		
5	0	0	50		
6	0	0	1050		

True portfolio value: 2617

Portfolio Yields

(Discount rate: 8.000%)

	Cash Flow				
Period	Bond A	Bond B	Bond C	Total Cash Flow	Discounted Cash Flow
1	40	35	50	125	
2	1040	35	50	1125	
3	0	35	50	85	
4	0	1035	50	1085	
5	0	0	50	50	
6	0	0	1050	1050	

True portfolio value: 2617

Portfolio Yields

	Cash Flow				(Discount rate: 8.000%)
Period	Bond A	Bond B	Bond C	Total Cash Flow	Discounted Cash Flow
1	40	35	50	125	115.74
2	1040	35	50	1125	964.51
3	0	35	50	85	67.48
4	0	1035	50	1085	797.51
5	0	0	50	50	34.03
6	0	0	1050	1050	661.68

True portfolio value: 2617

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True portfolio value: 2617 **Value: 2640.94**

Portfolio Yields

	Cash Flow				(Discount rate: 8.272%)
Period	Bond A	Bond B	Bond C	Total Cash Flow	Discounted Cash Flow
1	40	35	50	125	115.45
2	1040	35	50	1125	959.67
3	0	35	50	85	66.97
4	0	1035	50	1085	789.53
5	0	0	50	50	33.60
6	0	0	1050	1050	651.78

True portfolio value: 2617 **Value: 2617**

Yield to Maturity Useful for Fixed Rate But...

- As with computing **prices**, several characteristics come to our aid:
  1. Fixed cash flows
  2. Fixed discount rates
  3. Fixed maturity

- How can we use our **yield to maturity** framework to evaluate
  - bonds with embedded options?
  - bonds with floating rates?

Yield to Maturity of a Callable Bond

- Let's consider a bond that's **callable** at some point before maturity
  - Not callable *yet*
  - Symbolically, call this time T_C $(< T)$

- **Q: What's the challenge we face, even given T_C?**

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  - We don't know if the issuer will actually call!
  - Depends on market prices, which are uncertain

- **So what should we do?**

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  - Depends on market prices, which are uncertain

- **So what should we do? A: Worst case scenario — yield to call**

How do we calculate yield to call?

(Example: 5 year corporate with par $1000 that's callable in one year at par)

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- **Same as YTM!** Just replace:
  (1) "`periods`" $\rightarrow$ "`periods until callable`"
  (2) "`par value`" $\rightarrow$ "`strike price`"

- **Q: Based on our previous exercises, how should this affect yields?**

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- But wait... last time we said callability $\rightarrow$ lower prices. **How do we rationalize this?**
  - Prices $\downarrow$ if callable *given yields!*

Call Schedules

- So far been a bit imprecise about callable bonds
  - Assumption 1: Call price = par value
  - Assumption 2: Single call date in the future

  Reality: Callable bonds often have a *call schedule*

- Q: Suppose a schedule has multiple call dates. How might you expect call prices to vary over time?

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  Reality: Callable bonds often have a *call schedule*

- Q: Suppose a schedule has multiple call dates. How might you expect call prices to vary over time?
  - **A: They typically decline!**
  - Want to *discourage* early calls

Example 1: Call Schedules and Yield to Call, Discount Bond

- Let's return to a 5-year corporate bond with:
  - Annual coupon rate 8%, price $950, par value $1,000

Periods to call	Call Price	Yield to Call
1	$1,050	
4	$1,025	
8	$1,000	

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1	$1,050	9.48
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8	$1,000	

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4	$1,025	5.14
8	$1,000	4.44

- **Q: Which one is the "relevant" yield?**
  A: The lowest — "yield to worst" — is standard practice in industry

Example 2: Call Schedules and Yield to Call, Premium Bond

- Let's return to a 5-year corporate bond with:
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- Let's return to a 5-year corporate bond with:
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1	$1,050	1.379
4	$1,025	
8	$1,000	

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4	$1,025	2.866
8	$1,000	

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1	$1,050	1.379
4	$1,025	2.866
8	$1,000	3.192

- **Q: Which one is the "relevant" yield?**
  A: Yield to par! **Why?**

Key Assumption: Holding to Maturity!

- The yield to maturity is simply a *measure* of the return on a bond <u>*under the assumption it is held until it matures*</u>

- **But what if we don't hold the bond until maturity?**

$$\text{Return} =$$

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$$\text{Return} = \underbrace{\frac{C}{P}}_{\substack{\text{fixed} \\ \text{cash flow}}}$$

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- **But what if we don't hold the bond until maturity?**

$$\text{Return} = \underbrace{\frac{C}{P}}_{\substack{\text{fixed}\\ \text{cash flow}}} + \underbrace{\frac{\Delta P}{P}}_{\substack{\textbf{uncertain}\\ \textbf{cash flow!}}}$$

$\rightarrow$ Must **project** how a bond's value changes to understand <u>*intermittent*</u> returns

- **Next Time:** How do changes in yields affect bond returns?